Report

Real Time Adaptive Signal Conditioning System for Wide Dynamic Range Inputs

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From January 5, 2026 to January 31, 2026

Academic Year 2025–2026

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Acronym	**Full Meaning**	**Where Used**
BOM	Bill of Materials	Section IV.10
BJT	Bipolar Junction Transistor	Section IV.10 (BOM table)
CAD	Computer Aided Design	Section IV.9
CV	Control Voltage	Sections III, IV.1, IV.3, IV.5, IV.6, IV.9
DPDT	Double Pole Double Throw	Section IV.10 (BOM table)
GND	Ground	Sections IV.9, IV.10
ISET	Institut Supérieur des Études Technologiques	Title page, Chapter II
ISIS	Intelligent Schematic Input System (Proteus)	Section IV.9
LED	Light Emitting Diode	Sections IV.8, IV.9, V, VI
LMD	Licence Master Doctorate	Chapter II
NPN	Negative Positive Negative (transistor type)	Section IV.10 (BOM table)
Op Amp	Operational Amplifier	Sections III, IV, V
PCB	Printed Circuit Board	Sections IV.9, IV.10, V, VI
POT	Potentiometer	Section IV.6
RC	Resistor Capacitor	Sections IV.2, IV.7
SMD	Surface Mount Device	Section VI
STIC	Sciences et Technologies de l'Information et de la Communication	Title page
TND	Tunisian Dinar	Section IV.10 (Cost analysis)
TS	Tip Sleeve (audio connector type)	Section IV.10 (BOM table)
USD	United States Dollar	Section IV.10 (Cost analysis)
VC	Voltage Control	Sections III, IV.3
VCA	Voltage Controlled Amplifier	Sections III.2, III.5

Table 1: List of Acronyms

I. General Introduction

Variations in input signal amplitude can have a significant impact on the performance of analog signal processing systems, resulting in distortion, saturation, or inadequate signal levels for proper processing. Signal conditioning circuits are commonly employed to regulate and adjust signal levels in real time in order to address these issues. Such systems are widely used in audio electronics, instrumentation, and control applications where reliable and predictable signal behavior is required.

The objective of this project is the design and implementation of an analog signal conditioning system capable of dynamically adjusting signal amplitude in response to changes in input level. The system monitors the input signal, extracts amplitude information, and modifies the gain of the signal path accordingly to maintain a controlled output range. This approach preserves the essential characteristics of the original signal while ensuring dependable operation.

The proposed architecture is organized into functional blocks, comprising a core processing system for signal control and an optional block for additional amplification and visual indication.

The project follows a structured engineering methodology, beginning with schematic design and component selection, proceeding through cost and power consumption analysis, and concluding with printed circuit board (PCB) design for the physical realization of the system. The end result is a complete, manufacturable PCB solution that satisfies the original functional requirements and provides a solid foundation for further testing or expansion.

The entire project has been made accessible through a dedicated GitHub repository to ensure transparency, reproducibility, and accessibility. All design data, including schematics, PCB layouts, documentation, and related resources, are contained in this repository, enabling a thorough review of the project from its initial stages of development to its final implementation. GitHub's version control capabilities make it possible to track all updates and changes made during the design phase, providing a clear record of the project's evolution. This approach facilitates collaboration, future improvements, and independent verification of the work presented in this report.

GitHub

`https://github.com/GarablueX/Signal-Conditioning-System`

II. Host Institution and Internship Environment

2.1 The Host Institution's Overview

The internship was completed in the Electrical Engineering Department of the Higher Institute of Technological Studies of Sousse (ISET Sousse). With an emphasis on applied technology education, ISET Sousse is a public university that follows the LMD (Licence Master Doctorate) academic system. The institute prepares students for technical and industrial situations by combining classroom education with hands on training.

Academic instruction in signal processing, electrical systems, and electronics is offered by the Department of Electrical Engineering. Its teaching methodology places a strong emphasis on developing practical skills through internships, technical projects, and laboratory work. In this context, project based internships foster technical discipline and autonomy while enabling students to apply theoretical knowledge to structured engineering work.

Figure 1: ISET Sousse

2.2 The Role and Environment of the Internship

The internship took place in the Department of Electrical Engineering and was project based, lasting one month. This internship, as opposed to an industry placement, concentrated on the autonomous completion of an entire engineering design project in an academic environment.

Throughout this period, the role undertaken was that of a project based engineering intern. The objective was to manage the development process from the initial concept to the final technical documentation while designing and documenting a signal conditioning subsystem. The work required the application of analog electronics fundamentals, signal conditioning theory, and PCB design techniques.

The development procedure followed a planned progression. The first step consisted of establishing the system's objectives and determining its primary functional components. Schematic creation and iterative simulation were then used to refine circuit parameters and confirm anticipated behavior. After the design stabilized, CAD tools were used to create the PCB layout, paying close attention to design rule verification, routing strategy, and grounding considerations. The final step involved compiling the technical documentation, assessing cost and power consumption, and creating the bill of materials.

2.3 Project Requirements and Scope

The primary objective of the project was the development of a signal conditioning subsystem for incorporation into an audio dynamics control application. The system is required to process analog signals in order to produce appropriate control behavior based on amplitude characteristics.

The principal technical goal was to determine a logical system architecture that included voltage control, peak detection, threshold comparison, adjustable temporal behavior, and voltage protection mechanisms. The design operates from dual $\pm 12\,\mathrm{V}$ supply rails, and parameters were made adjustable to facilitate future experimental tuning and evaluation.

Due to the internship's one month timeframe, the scope of work was restricted to the design and simulation phase. No physical fabrication, assembly, or laboratory measurements were carried out during this period. Accordingly, validation was performed through circuit modeling, analytical computations, and electronic design rule verification.

2.4 Internship Deliverables

The final deliverables at the conclusion of the internship comprised the following: system architecture definition, complete schematic design, simulation results demonstrating functional behavior, PCB layout created in KiCad, a detailed bill of materials with cost estimation, power consumption analysis, and organized technical documentation maintained in a version controlled repository.

III. Detailed explanation of how the system works

This chapter walks through how the system works, block by block. Each functional stage is explained individually to show its role in the overall signal path, from input to output.

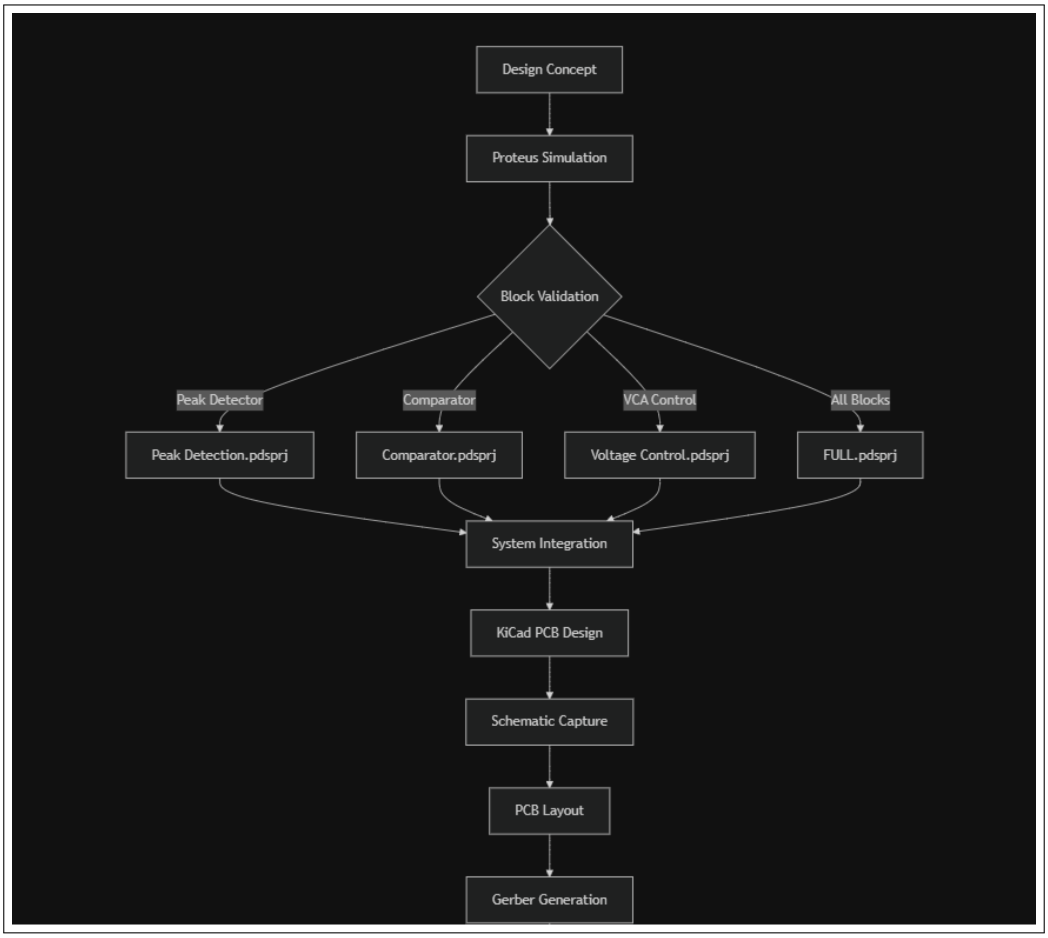

Figure 2: Design Workflow

1 Signal Input

The system receives an analog input signal whose amplitude may vary significantly over time.

2 Voltage Control

The input signal is fed directly into a Voltage Controlled Amplifier. This block applies a gain to the signal, where the gain value is determined by an external control voltage. The VCA is responsible for increasing or decreasing the signal amplitude in real time.

At this stage: The signal path remains continuous No frequency modification is intended Only amplitude scaling is applied

3 Envelope Detector

In parallel with the main signal path, the input signal is routed to an Envelope Detector. The purpose of this block is to extract a representation of the signal's amplitude (or level), independent of its frequency content. This is typically achieved through rectification followed by low pass filtering, producing a slowly varying control signal. This block provides the system with real time information about the strength of the input signal.

4 Signal Comparator

The output of the Envelope Detector is fed into a Signal Comparator, where it is compared against a predefined reference threshold level. If the detected signal level is below the threshold, no gain reduction is required. If the detected signal level exceeds the threshold, the comparator generates a control signal indicating that gain reduction should be applied. This block implements the decision logic of the system.

5 Control Signal Generation

The output of the Signal Comparator is converted into a control voltage that determines how much gain reduction is applied by the VCA. The magnitude and timing of this control signal govern: How quickly the system responds to increases in input level How smoothly the gain returns to normal when the input level decreases This behavior defines the system's dynamic response characteristics.

6 Signal Output

The processed signal exits the system through the output block. As a result of the adaptive gain control, the output signal exhibits a reduced dynamic range compared to the input, remaining within a controlled amplitude range even when the input varies significantly.

7 System block illustration

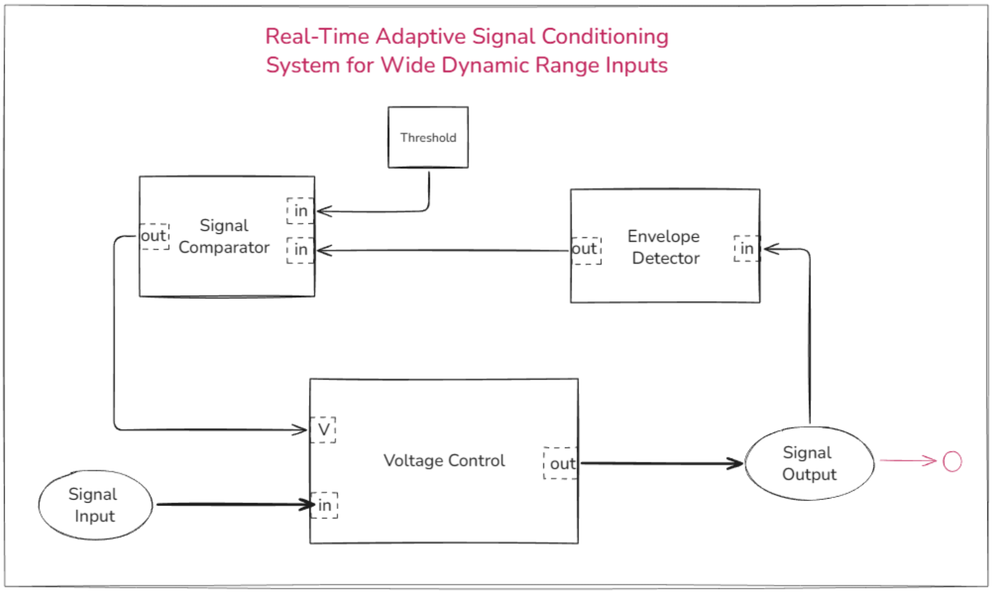

Figure 3: Block diagram

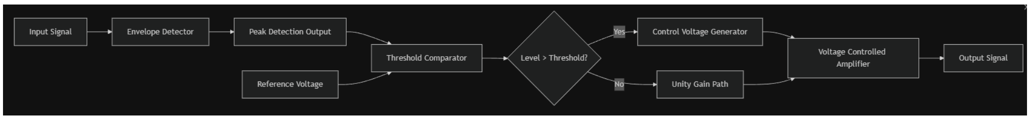

Figure 4: Topology

IV. Deployment

This chapter covers the practical side how each block was actually built, what methods were considered, and why certain choices were made. Schematics and simulation results are included for each stage.

1 Voltage Control

Available methods:

- Multi variable resistor low pass filtering
- Diode ladder filtering

Chosen Method: Diode ladder filtering this method consists of combining multiple 1N4148 diodes in series, which act as a variable resistance based on the control voltage applied to the input of the CV (V). This method causes less distortion and provides greater control over the CV. It is also generally more cost effective and easier to assemble.

1.1 functioning

The selected technique involves utilizing multiple diodes as variable resistors. It functions by supplying symmetric voltages to the ends of the diode ladder and a non inverted

buffered input at the center.
This is the fundamental principle behind the diode ladder. If the symmetric voltages are zero, the output signal is passed through unchanged from the middle of the ladder. If the symmetric voltages are non zero, current flows from the positive rail to the negative rail, simultaneously diverting some of the current from the input signal. As a result, the output signal retains the same waveform as the input but with a reduced amplitude.
Therefore, as the symmetric voltages increase, the input signal is attenuated further.

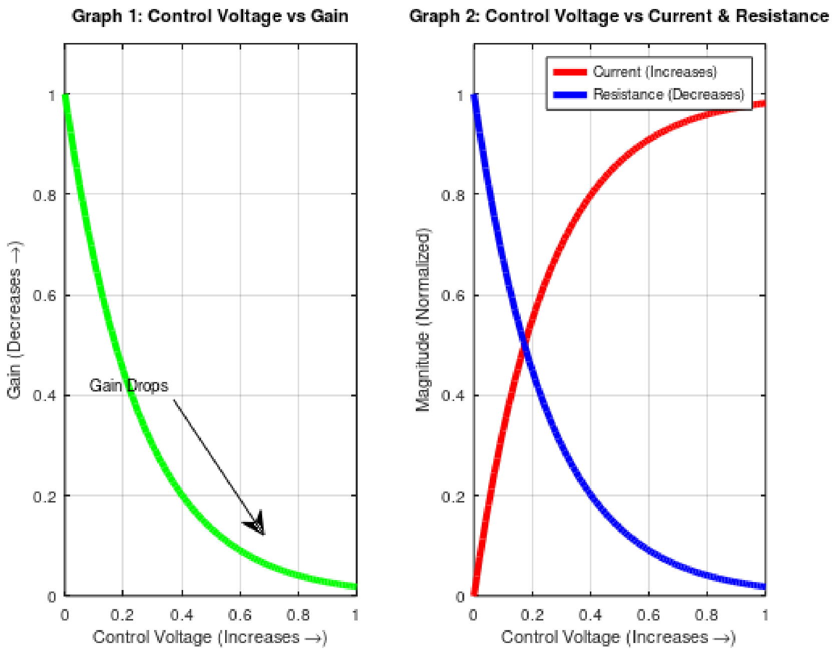

Figure 5: Diode Ladder Control Characteristics

1.2 Schematic

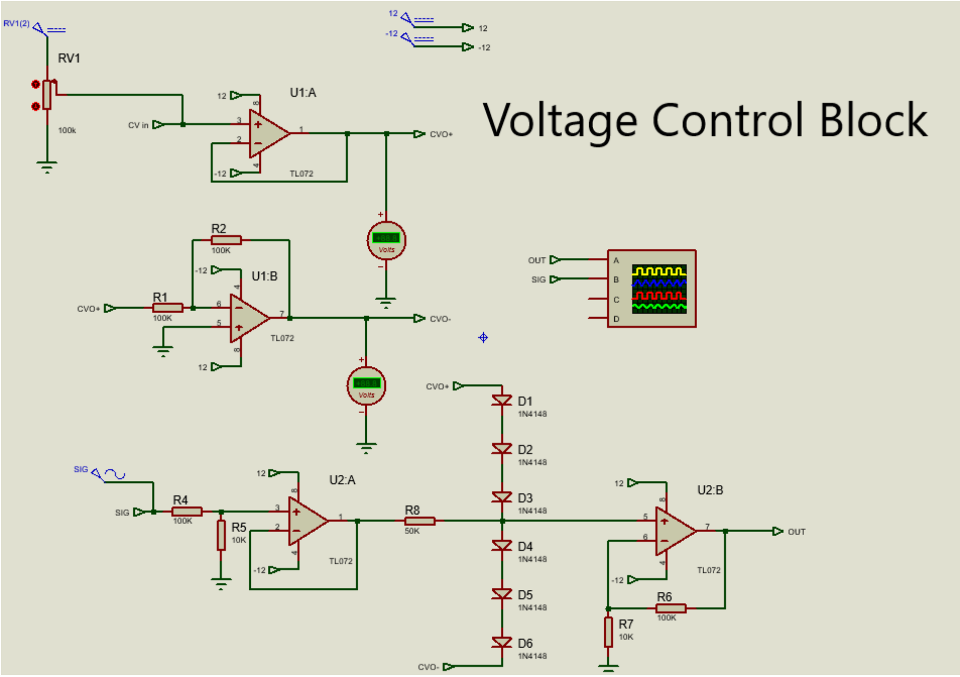

Figure 6: Voltage Control Block schematic

1.3 Simulation

CV = 0 → Gain = 1 CV = 2 → Gain < 1

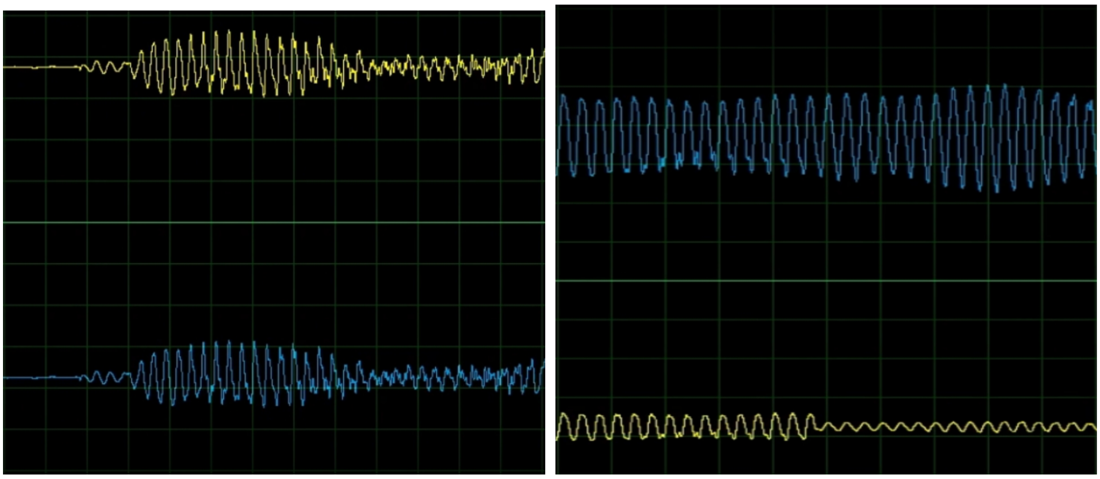

Figure 7: Simulation 1 and Simulation 2

2 Peak Detector

Available Methods

There are several approaches for peak detection in analog circuits, including the following:

- Diode capacitor circuit
- Diode capacitor amplifier circuit
- Diode capacitor RC coupling circuit

Chosen Method

The diode capacitor RC coupling technique was selected due to its simple circuit topology, high efficiency, and fast response time.

2.1 functioning

A peak detector can be constructed by connecting a buffer op amp through a diode to an RC coupling network.
When the input signal increases, the capacitor charges through the diode. When the input signal decreases, the diode blocks reverse current flow, and the charge stored in the capacitor drains through the resistor to ground.
This arrangement allows the peak amplitude to be detected at the output of the resistor.

2.2 Schematic

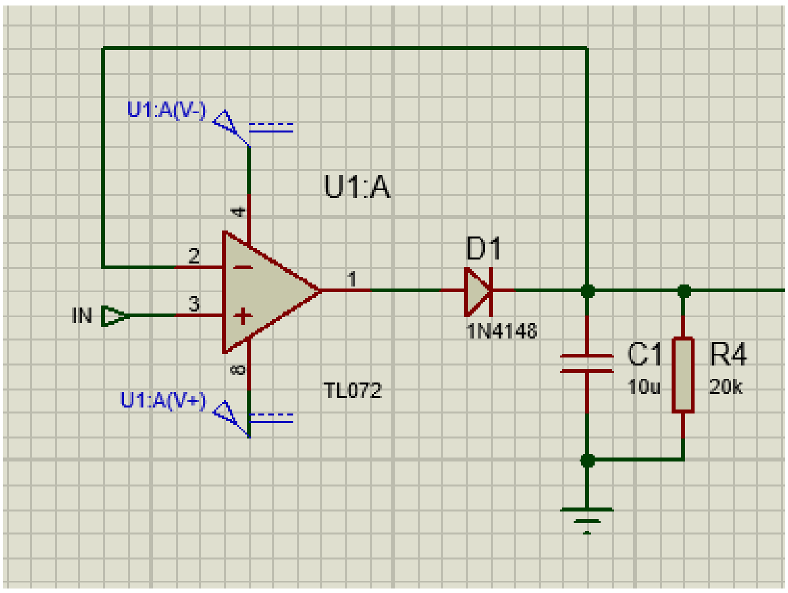

Figure 8: Schematic Peak detector

2.3 Simulation

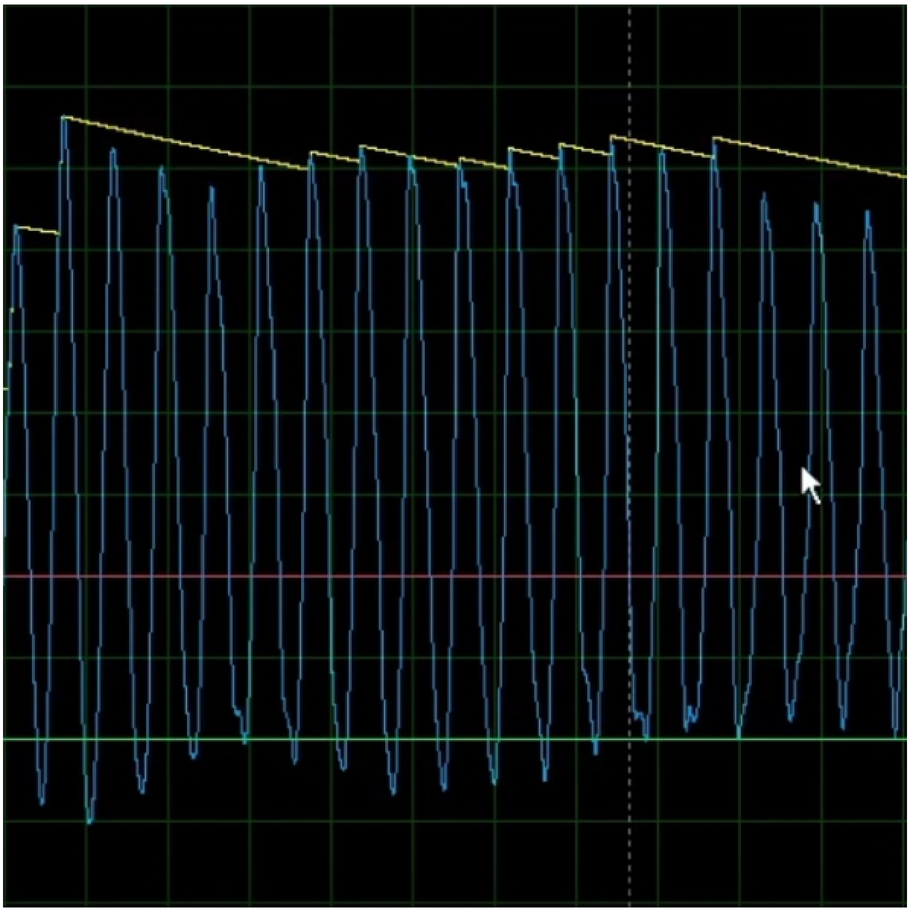

Figure 9: Simulation Peak detector

Note:
Due to the orientation of the diodes in the control voltage block (described in the preceding section), negative peaks are not required. In any case, since all signals are expected to be approximately sinusoidal, the negative half cycles are not needed.

3 Signal Comparator Block

3.1 functioning

The operation of the comparator block is straightforward: it is implemented using a differential amplifier with unity gain.
The threshold voltage is connected to the amplifier's non inverting input. A negative output indicates that the input signal is below the threshold, meaning no gain reduction is required. A positive output indicates that the input signal exceeds the threshold, signaling that gain reduction should be applied.
The output of the comparator block serves as the input to the VC block.
Since the maximum input is 12 volts peak to peak, the threshold element is a variable voltage that ranges between 0 and 6 volts.

3.2 Schematic

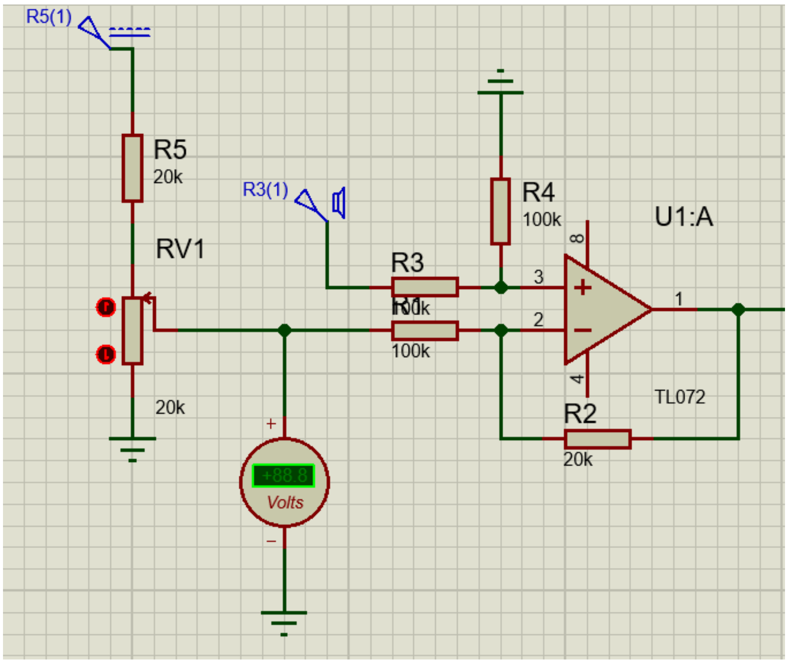

Figure 10: Comparator Block

4 Assembling

The assembly process is straightforward, consisting of combining the previously described blocks and fine tuning the wiring and input/output connections.

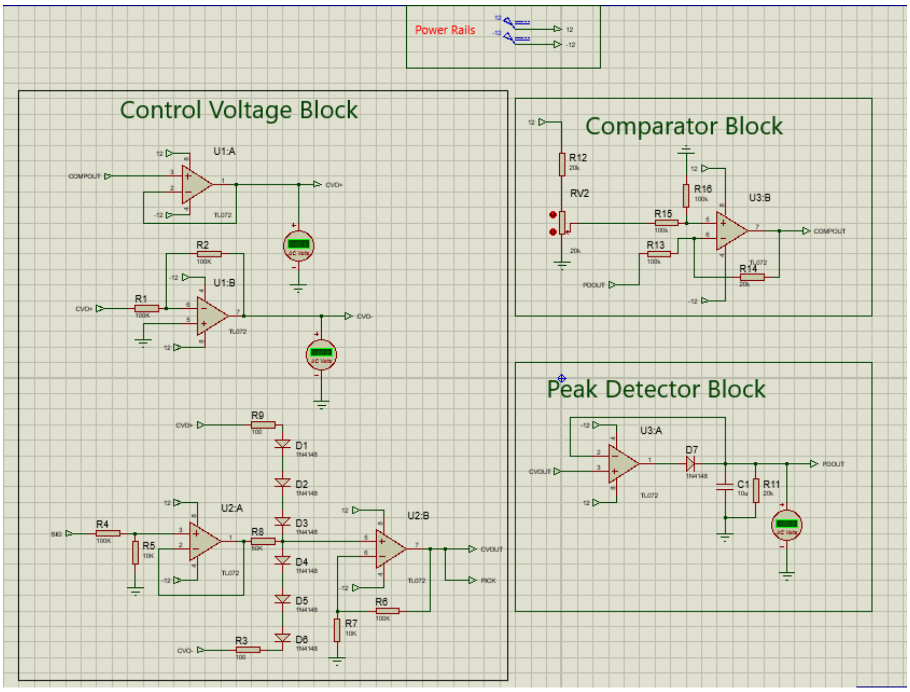

Figure 11: Assembled Blocks

5 Initial Performance Analysis

Through simulation testing and analysis, several issues were identified. While the system does not fail entirely, its behavior deviates from the desired performance in the following ways:
First, the control voltage inputs (which correspond to the comparator outputs) remain positive even when the threshold is set to 0 volts. As a result, the output signal from the CV block is always attenuated by an unwanted factor.
Additionally, it was observed that when the input difference exceeds approximately 2 volts, the output is completely silenced. Furthermore, the output signal exhibits noticeable distortion under these conditions.

6 Proposed Solutions

The primary issue causing the output to be silenced is that the CV input becomes excessively high (approximately 2.5 volts), which fully opens the diodes and drains the output signal entirely to ground.
The proposed solution is to insert a voltage divider with a ratio of 1/2 between the comparator output and the CV input.
Through experimentation, it was determined that using a 30 kΩ resistor and a 10 kΩ resistor with a variable potentiometer yields the best results for the desired operation.

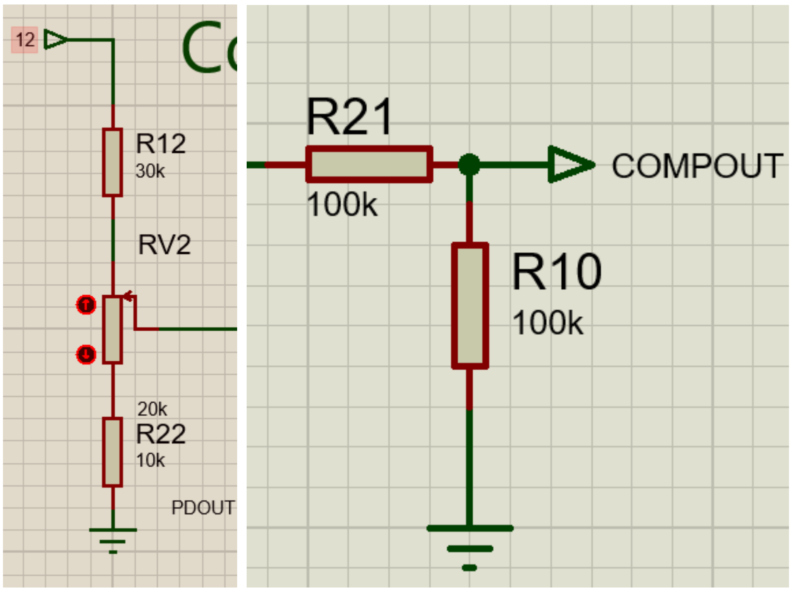

Figure 12: Refined POT and Comparator Output

7 Adding features

7.1 Attack

Definition
Attack refers to how quickly the system begins reducing gain after a signal exceeds the

defined threshold. In other words, it determines how long the system waits before attenuating the input.

Explanation

During the testing and debugging phase of the project, a simple and effective method for implementing the attack control was identified. This approach is based on controlling how quickly the peak detector capacitor responds to incoming signal peaks.

By slowing the charging rate of the capacitor, the peak detector responds more gradually, which in turn reduces how quickly peak information is conveyed to the comparator.

As a result, replacing the fixed resistor between the diode and the capacitor with a variable resistor provides a practical means of implementing the attack feature. In this design, a 100 kΩ potentiometer was used for this purpose.

7.2 Release

Definition

The release time is the duration required for the control system to remove gain reduction and return to its nominal state after the input signal level drops below the threshold. It governs how the system recovers following attenuation.

Explanation

The release time is implemented using a simple RC based approach. The release time constant is defined as $\tau = RC$, where 10 μF is the fixed value of the capacitor.

The release time is controlled by varying the resistance in the discharge path using a 100 kΩ potentiometer.

To prevent the discharge path from being shorted to ground when the potentiometer is set to its minimum value, a 5 kΩ fixed resistor is placed in series with the potentiometer. This ensures a non zero minimum resistance and provides safe, stable operation.

As a result, the release time constant is adjustable over a range defined by:

$$\tau \in [5\,\mathrm{k\Omega} \times 10\,\mu\mathrm{F},\ 105\,\mathrm{k\Omega} \times 10\,\mu\mathrm{F}] = [50\,\mathrm{ms},\ 1.05\,\mathrm{s}]$$

Schematics After adjustment

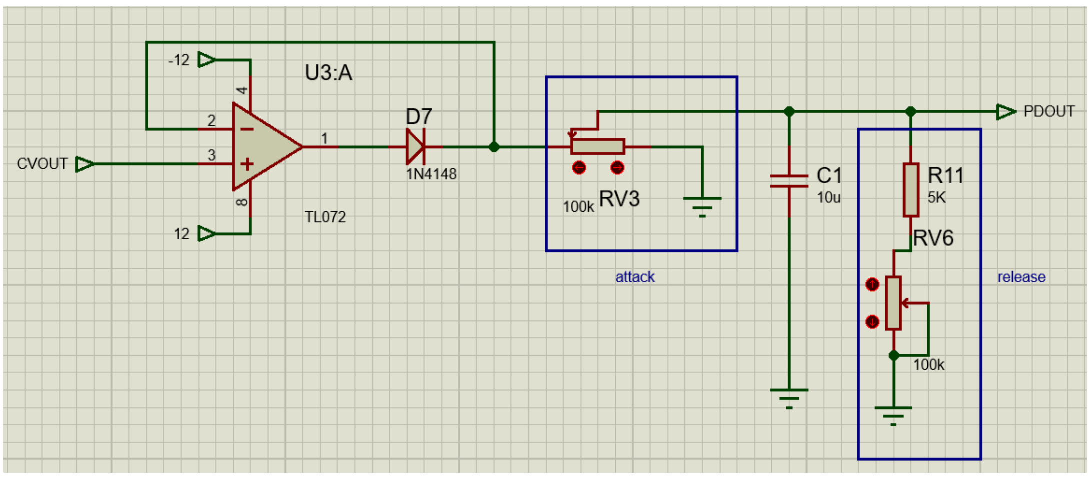

Figure 13: Attack Release

7.3 Ratio

Definition
The compression ratio defines the degree to which the system reduces the output signal level relative to the amount by which the input level exceeds the defined threshold.
Explanation
The ratio determines the relationship between input level exceedance and the resulting gain reduction applied by the system. In the proposed design, the ratio is implemented by scaling the comparator output using a variable voltage divider.
By attenuating the control voltage that drives the gain control stage, the system adjusts how aggressively gain is reduced once the threshold is exceeded. A higher divider output results in stronger gain reduction (higher ratio), while a lower divider output produces gentler compression behavior (lower ratio).
Schematics After Adjustment

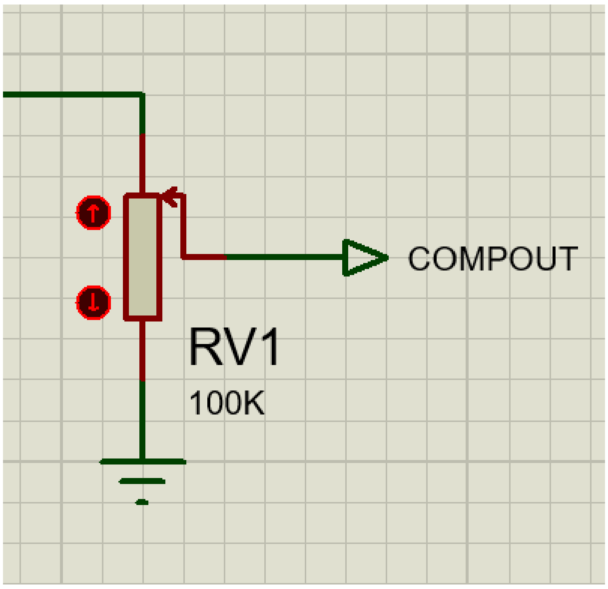

Figure 14: Ratio control circuit after adjustment

7.4 Gain Amplifier

Definition
An optional gain amplifier stage is included to compensate for cases in which the input signal level is insufficient. Two modes of operation are available: the output either remains unchanged or is amplified using a variable resistor.
Explanation
Adding a non inverting amplifier with variable gain, controlled by a variable resistor in series with a fixed one, provides a straightforward implementation of the gain feature.
As a result, this stage functions solely as an amplifier and cannot operate as an attenuator.

Schematics After Adjustment

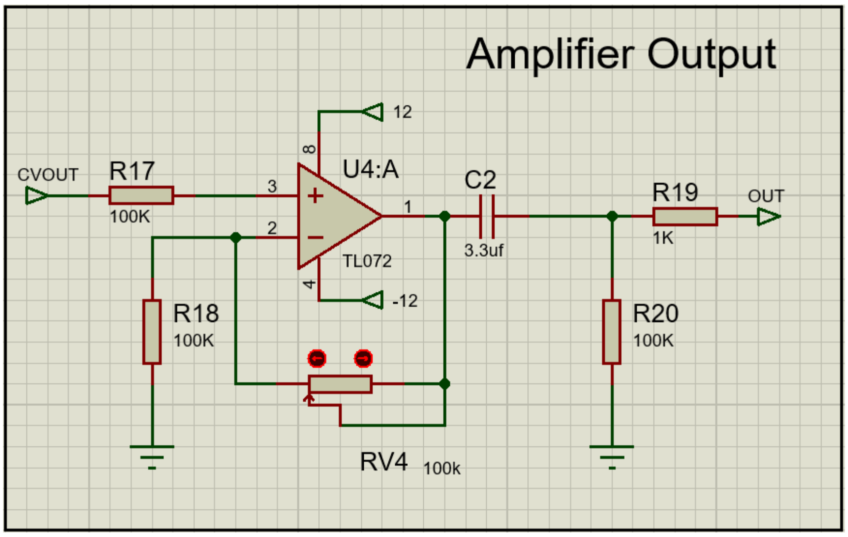

Figure 15: Gain amplifier circuit

8 Design Refinement & Additional Features

Based on observations during testing, the existing circuit's use of a negative threshold to force the signal to zero does not result in smooth attenuation. Instead, it introduces an abrupt cutoff point that may be undesirable for the user.

In addition, an LED indication system specifically an LED follower for basic visual monitoring of the output was identified as a valuable feature to include.

It was also determined that, given the current scope of the project, the amplifier and LED visualizer blocks should be separated from the core system as optional additions. The core system had become increasingly complex and costly, and these two blocks are not essential to its primary function. Accordingly, they are designed to operate as subsystems to the core, and their inclusion is entirely optional.

Since the threshold is set at 6 volts, an input clamping mechanism was added to limit the maximum input to 6 volts (12 volts peak to peak), ensuring that the system operates without distortion or risk of component damage.

8.1 LED Follower

Explanation

The LED follower block is implemented using a set of 2N2222 transistors. Since the peak detector is already operational, its output can be amplified and used to drive individual LEDs at discrete levels by means of resistive voltage dividers for each threshold level.

Schematics After adjustment

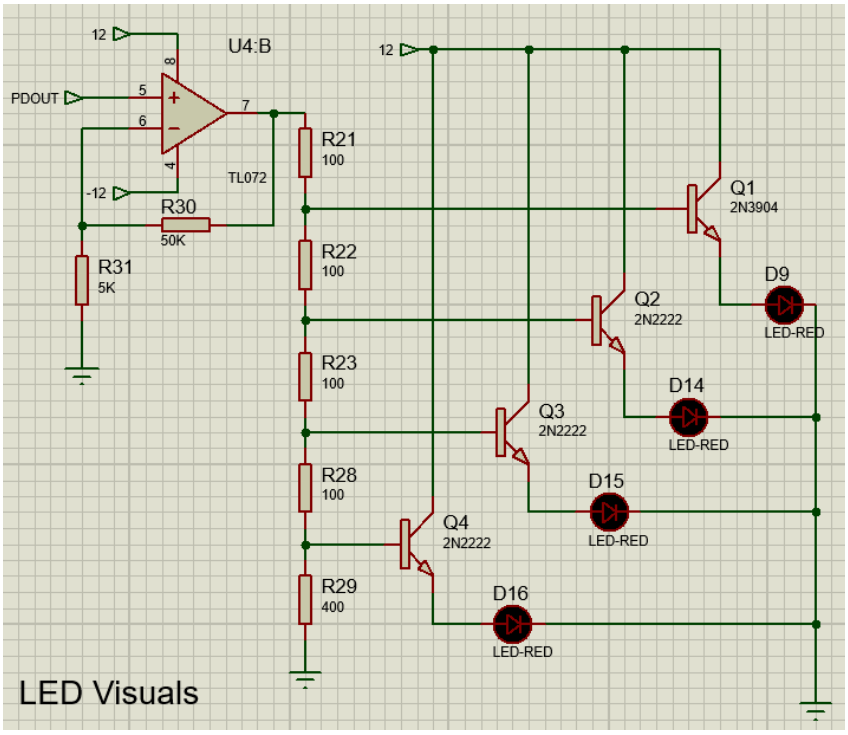

Figure 16: LED Follower circuit

8.2 Attenuation problem

This issue was resolved by a simple modification: replacing the linear potentiometer with a logarithmic one.

8.3 Maximum input

Since the threshold is approximately 6 volts, a Zener diode clamp was added to the system's input.
After a brief component search, it was determined that using two 1N5232B Zener diodes will clamp the input to a maximum of approximately 6.2 volts. This is acceptable for the intended application and ensures proper system operation within the designed voltage range.
Schematics After adjustment

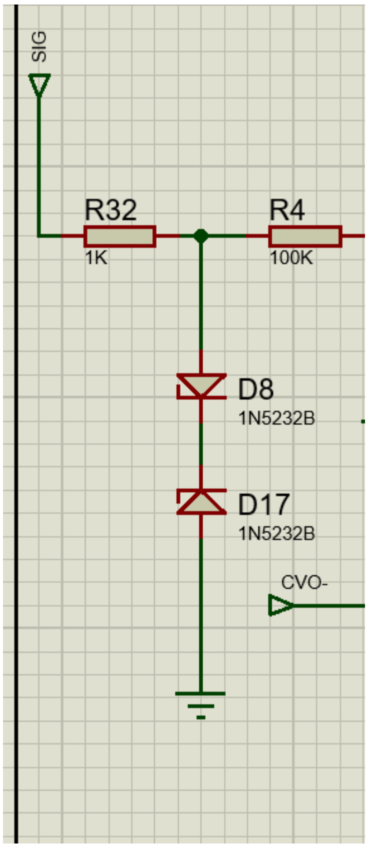

Figure 17: Input Clamping

9 Changing the CAD

Since the CAD program previously used (PROTEUS / ISIS from Labcenter) is a proprietary, paid application, the decision was made to migrate the project to an open source platform to ensure accessibility for the wider community. For this reason, KiCad was selected as the replacement, being the most widely used open source electronics design automation tool.

9.1 Changes

As this represents the final revision of the schematics, a number of modifications were made before proceeding to the PCB layout:

- All inputs are now configured to receive external signals.
- Three connectors were added for +12 V, -12 V power, and ground.
- A power indicator LED connected to +12 V was added to confirm that the system is powered on.
- Two audio jacks were added: one for input and one for output.
- A switch was added to alternate between the amplified output and the raw output, since the amplifier sub block is optional. If the sub amplifier system is not in use, the unused switch pin can be tied to ground.
- Since the LED follower and amplifier blocks are optional subsystems, two pin headers were added to facilitate connection between them and the core system using jumper wires. These pin headers are connected to the peak detector input and the CV output pins.
- Since the core system requires six amplifiers in total, one TL074 quad amplifier and one TL072 dual amplifier were used instead of three TL072 dual amplifiers, reducing both cost and board space. The sub blocks (LED follower and amplifier), which are grouped together, are powered by a single TL072 dual amplifier.
- Two additional pins were added for supplying +12 V and -12 V to the standalone subsystems.

9.2 Schematics after changes

Control Voltage Block

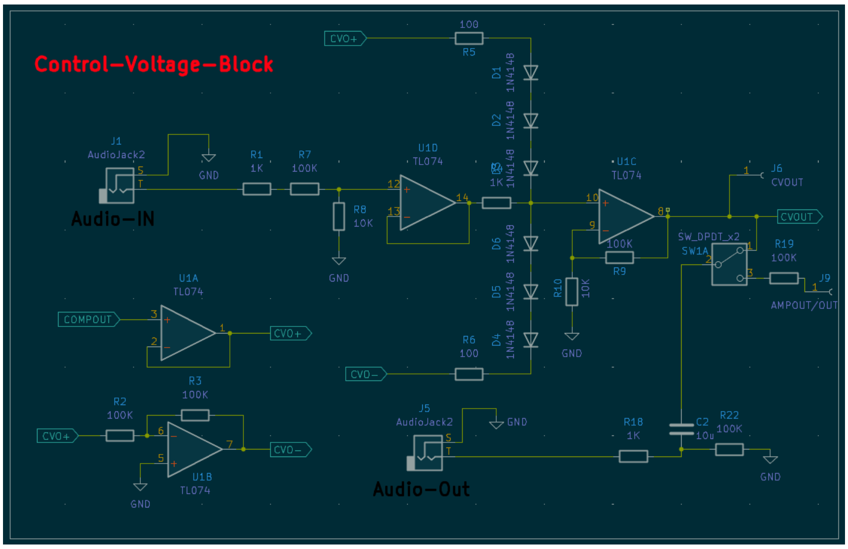

Figure 18: CV with KiCad

Peak Detector Block

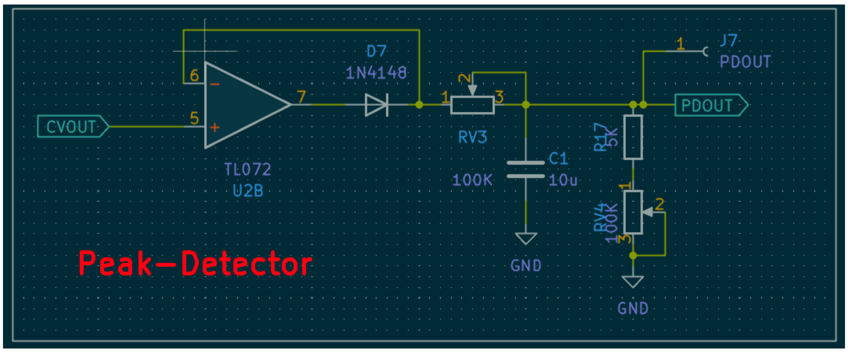

Figure 19: Peak Detector with KiCad

Comparator

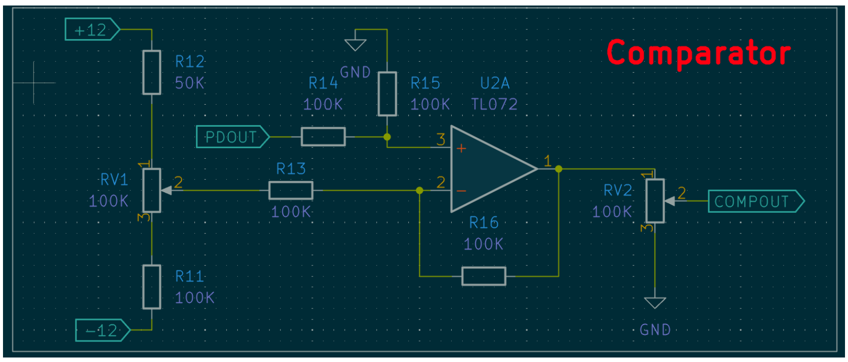

Figure 20: Comparator with KiCad

Power Rails

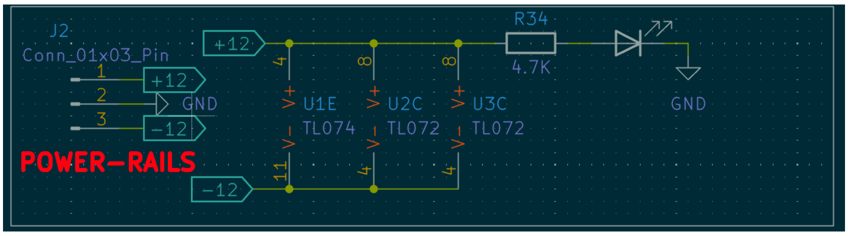

Figure 21: Power Rails with KiCad

Amplifier

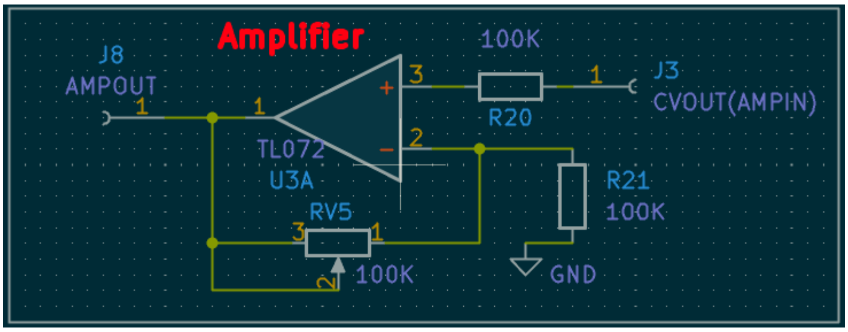

Figure 22: amplifier with KiCad

LED Follower Block

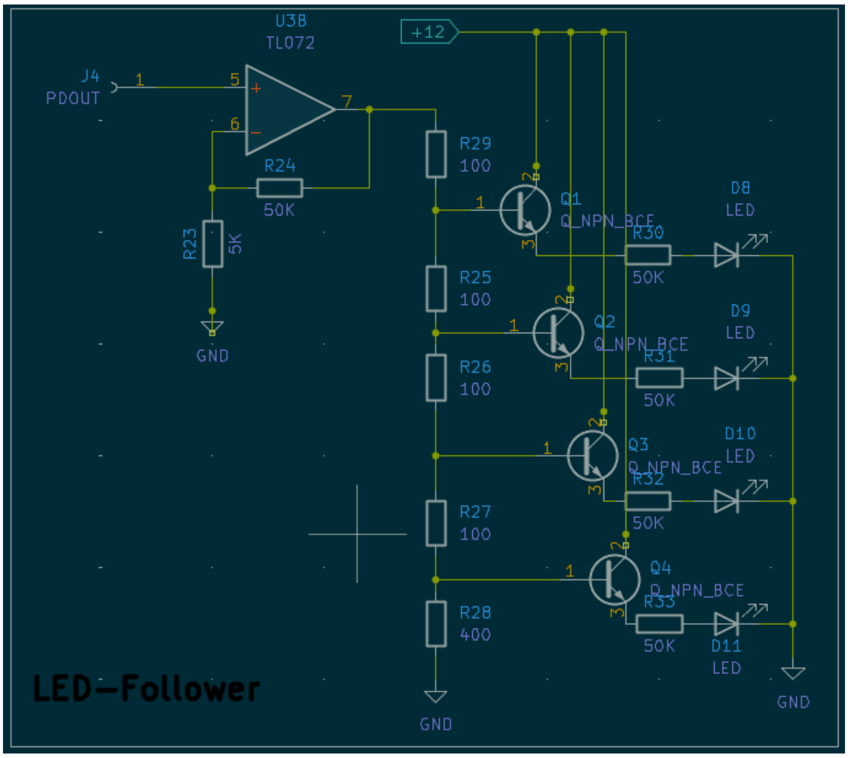

Figure 23: LED Follower Block with KiCad

Full schematics

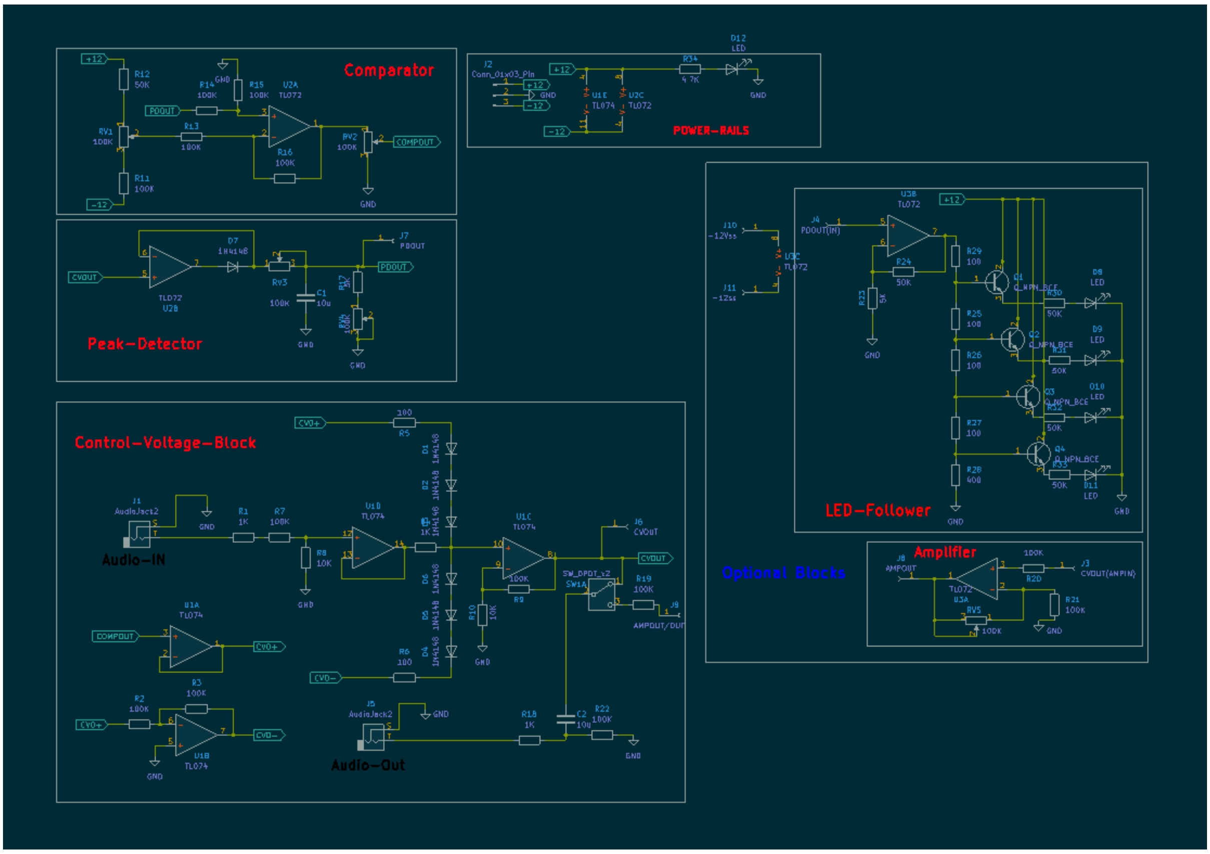

Figure 24: Full schematics with KiCad

The following diagram provides a high level overview of the signal flow through the finalized system.

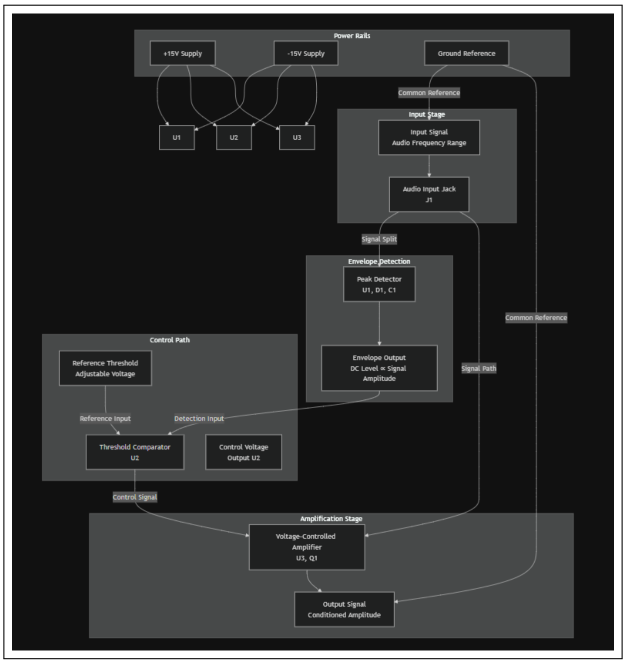

Figure 25: System Signal Flow with Component References

10 Preliminary Bill of Materials and Cost Analysis

10.1 Bill of Materials

Reference(s)	Qty	Component Description	Value / Part
C1, C2	2	Electrolytic capacitor	10 μF
D1–D7	7	Signal diode	1N4148
D8–D12	5	Indicator LED	3 mm LED
J1, J5	2	Mono audio jack (TS)	6.35 mm
J2	1	Power connector	3 pin (+12 V / GND / −12 V)
J3–J11	7	Pin header / test point	1×1
Q1–Q4	4	NPN transistor	General purpose BJT
R1, R4, R18	3	Resistor	1 kΩ
R2, R3, R7, R9, R11, R13–R16, R19–R22	13	Resistor	100 kΩ
R5, R6, R25–R27, R29	6	Resistor	100 Ω
R8, R10	2	Resistor	10 kΩ
R12, R24	2	Resistor	50 kΩ
R17, R23	2	Resistor	5 kΩ
R28	1	Resistor	400 Ω
R34, R30–R33	3	Resistor (LED limiter)	4.7 kΩ
RV1–RV5	5	Trimmer potentiometer	100 kΩ
SW1	1	Toggle switch	DPDT
U1	1	Operational amplifier	TL074
U2, U3	2	Operational amplifier	TL072

Table 2: Bill of Materials

The bill of materials was generated from the finalized schematic in KiCad. Package footprints and PCB specific details are omitted, as the BOM represents the schematic level design prior to PCB layout and routing.

10.2 Cost Analysis

Item	Qty	Unit (USD)	Subtotal (USD) / (TND)
TL074 (quad op amp)	1	1.31	1.31 / 3.82
TL072 (dual op amp)	2	1.04	2.08 / 6.05
Trimmer potentiometer Bourns 3386P	5	2.20	11.00 / 32.02
Neutrik NJ2FD V mono jack (TS)	2	2.05	4.10 / 11.93
Phoenix Contact 3 pos power connector (1836309)	1	1.23	1.23 / 3.58
C&K JS202011CQN DPDT switch	1	0.74	0.74 / 2.15
2N3904 (or similar) NPN transistor	4	0.15	0.60 / 1.75
1N4148 diode	7	0.10	0.70 / 2.04
3 mm LED (indicator)	5	0.40	2.00 / 5.82
10 μF electrolytic capacitor	2	0.21	0.42 / 1.22
Resistors (axial, ¼W)	34	0.01*	0.34 / 0.99
1×1 pin headers / test pins	7	0.05*	0.35 / 1.02
Estimated total (components only):			$24.87 72.38 TND

Table 3: Estimated Component Cost Analysis

The cost estimation is based on low quantity distributor pricing for representative parts. Passive components (resistors and basic headers) are costed using typical bulk pricing. This estimate excludes PCB fabrication, shipping, taxes, and assembly costs.

V. Power Consumption and Dissipation Analysis

Supply rails The circuit is powered from a symmetric dual supply: $V_+ = +12$ V $V_= -$ 12 V

Total power drawn from the supply is:

$$P_{\text{TOTAL}} = (12 \cdot I_+) + (12 \cdot I_)$$

1 Operational amplifier quiescent current (typical)

The TL07x family datasheets specify a typical supply current of approximately 1.4 mA per amplifier (no load, $T_A = 25°\text{C}$).

The design includes: TL074 (quad): 4 amplifiers Two TL072 (dual + dual): 4 amplifiers total

Total number of amplifiers:

$$N = 4 + 2 \cdot 2 = 8$$

Total quiescent current for all amplifiers:

$$I_{\text{opamp,T}} \approx N \cdot 1.4 \text{ mA} = 8 \cdot 1.4 \text{ mA} = 11.2 \text{ mA}$$

For a symmetric supply, a reasonable approximation is that the quiescent current is split equally between the rails:

$$I_{\text{opamp,+}} \approx I_{\text{opamp,-}} \approx \frac{11.2}{2} = 5.6 \text{ mA}$$

2 Indicator LED current and resistor dissipation

Each indicator LED is powered from +12 V with a 4.7 kΩ series resistor. Assuming a typical LED forward voltage of $V_f \approx 2.0$ V, the LED current when on is:

$$I_{\text{LED}} = \frac{12V_f}{R} = \frac{122}{4700} = 2.13 \text{ mA}$$

Worst case LED condition (all 5 LEDs on simultaneously):

$$I_{\text{LED,total}} = 5 \cdot 2.13 \text{ mA} = 10.65 \text{ mA}$$

Resistor power dissipation per LED:

$$P_R = I^2 \cdot R = (0.00213)^2 \cdot 4700 \approx 0.021 \text{ W}$$

Therefore, standard 0.25 W resistors provide a large safety margin.

3 Rail current estimates

Worst case (all LEDs on):

$$I_+ \approx I_{\text{opamp,+}} + I_{\text{LED,total}} = 5.6 + 10.65 = 16.25 \text{ mA}$$

4 Total power drawn from the supply

$$P_{\mathrm{TOTAL}} = 12 \cdot (I_{+} + I_{)}) = 12 \cdot (0.01875 + 0.0081) = 12 \cdot 0.02685 = 0.322\ \mathrm{W}$$

5 Conclusion

The circuit is low power by design. With a typical TL07x quiescent current of approximately 1.4 mA per amplifier and worst case LED activity, the total power drawn from a ±12 V supply is approximately 0.32 W.
The maximum resistor dissipation in the LED current limiting resistors is approximately 0.021 W, which is well below the standard 0.25 W rating.
As a result, no special thermal management measures are required for normal operation, and the selected component ratings provide adequate safety margins.

VI. Layout and Routing

Introduction
Following the validation of the schematic design and the completion of the preliminary bill of materials, cost estimation, and power consumption analysis, the next step in the design process is to physically realize the circuit in the form of a printed circuit board (PCB). This step converts the logical schematic into a manufacturable layout while maintaining signal integrity and ensuring reliable operation. Because the circuit processes analog, audio frequency signals, particular attention is given to component placement, grounding strategy, and routing order. Manual routing is preferred over automatic routing to limit noise coupling and enable controlled placement of sensitive signal paths. The following sections describe the layout strategy and design considerations employed during the PCB creation process.

1 PCB Design Without Optional Blocks

The system follows a modular architecture, as the PCB design incorporates several functional blocks beyond the essential signal processing path. Initially, a PCB dedicated exclusively to the core system was created, containing only the minimum circuitry required for full and correct operation of the primary function. This approach ensures that the system's core capabilities remain accessible and functional independently of the auxiliary blocks. Users are therefore not required to install or purchase the complete system if they do not need all the features. Accordingly, the LED follower stage and the additional amplification block which may be added later as optional extensions are excluded from the initial PCB design phase, which focuses solely on the core circuit. A filled ground plane serves as a common reference throughout the PCB. Routing was performed on both the top and bottom layers to facilitate trace placement. Power traces for the ± 12 V supply were routed with a width of 0.5 mm, while other signal traces used a standard width of 0.2 mm. The audio input and output connectors were positioned on opposite sides of the board for improved accessibility and organization, and the potentiometers were grouped together. Dedicated SMD test point pads were added for the $+12$ V, 12 V, CVOUT, and COMPOUT nodes to facilitate debugging and measurement as needed.

1.1 Full board

The following figure shows the complete PCB layout of the core system. It illustrates the final component placement and routing strategy used in the design, including the arrangement of functional blocks, power distribution, and grounding approach.

The design maintains a clear and organized signal flow appropriate for reliable operation, while respecting the mechanical constraints imposed by user interface elements such as potentiometers and audio connectors.

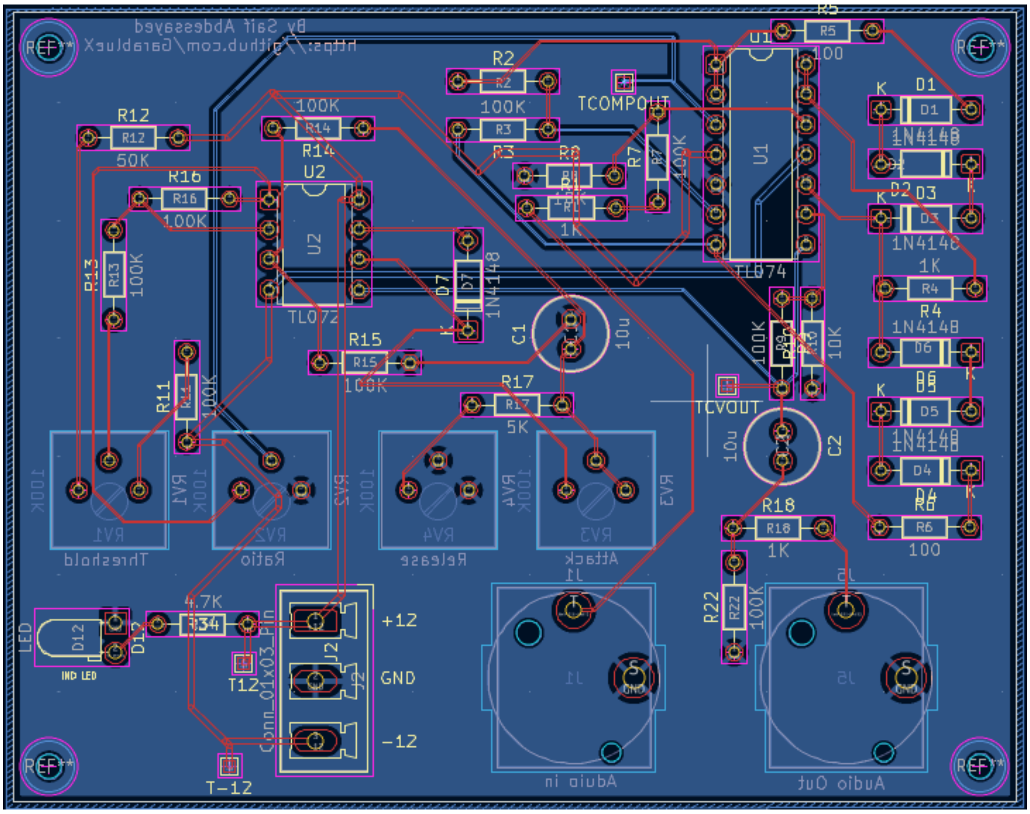

Figure 26: Full Layout

1.2 Top copper

The top copper layer was used primarily for signal and power routing. The ±12 V supply rails were routed with wider trace widths to improve reliability and minimize noise, while critical analog paths and op amp feedback connections were kept short and direct.

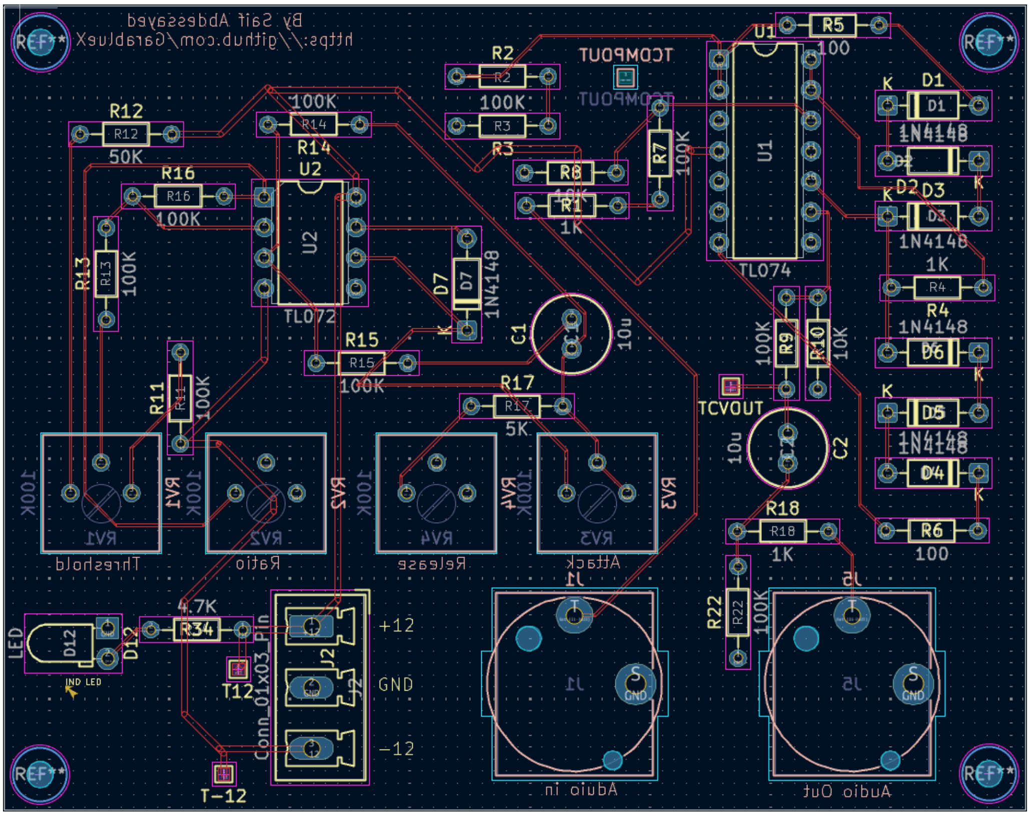

Figure 27: Top copper

1.3 Bottom Copper

The bottom copper layer was used predominantly as a continuous ground plane across the entire PCB. This ground plane provides a low impedance common reference for all circuit blocks, improving signal integrity and reducing noise coupling both of which are critical considerations for analog and audio frequency signals. To preserve the continuity of the ground plane, only minimal routing was performed on the bottom layer for necessary interconnections and vias. Thermal reliefs were used to maintain effective ground connectivity while ensuring reliable soldering of through hole components.

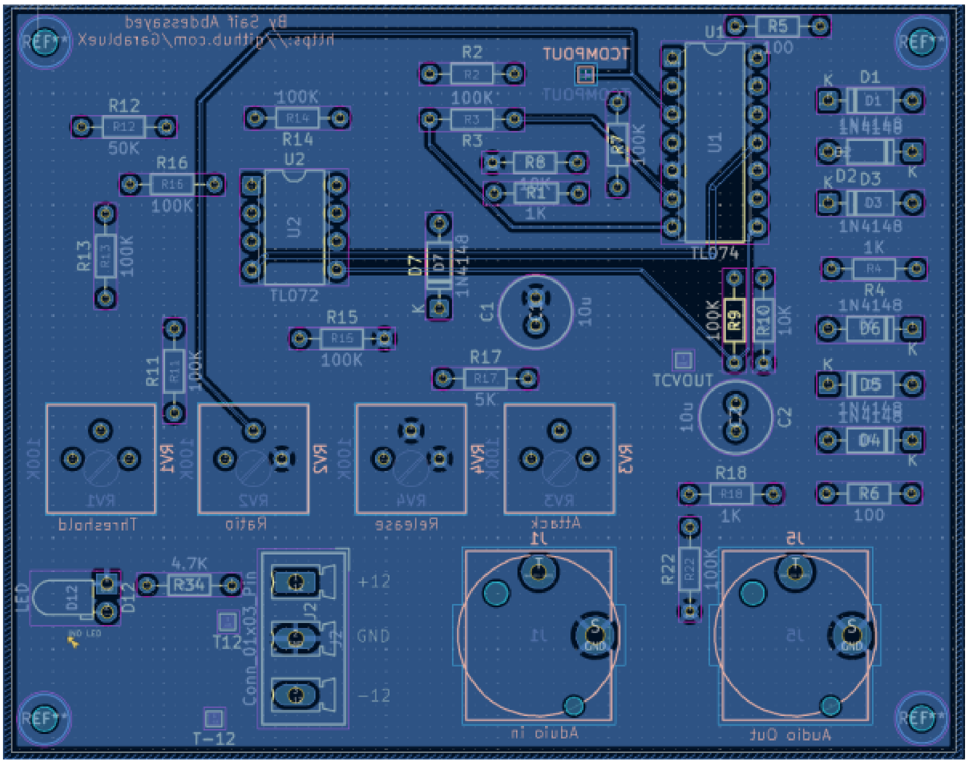

Figure 28: Bottom Copper

1.4 3D View

A three dimensional (3D) view of the PCB was generated to verify the mechanical integrity of the design and to validate component placement prior to fabrication. This view enables visual inspection of component heights, connector alignment, and overall board geometry, confirming that all mechanical elements fit within the board outline without interference. The 3D representation also verifies the accessibility of user interface components such as potentiometers, audio connectors, and switches, providing confidence that the PCB can be assembled and integrated into its intended enclosure.

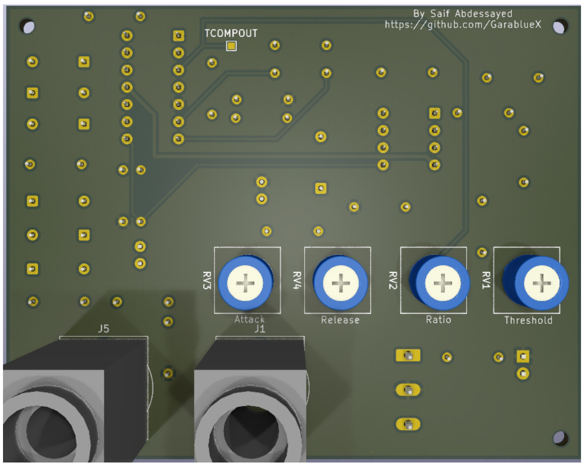

Figure 29: Front Face Core

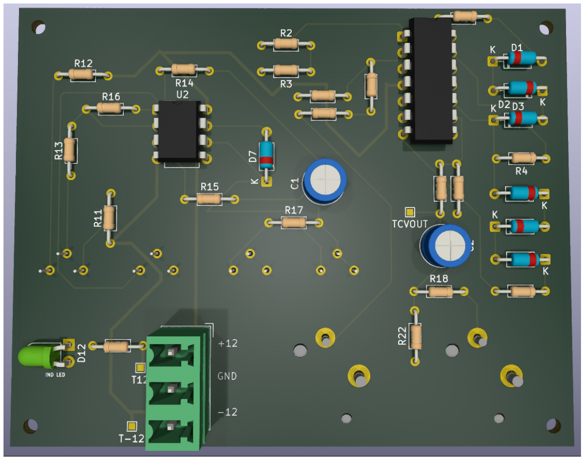

Figure 30: Back Face Core

2. PCB Design With Optional Blocks

In addition to the core system PCB, a second layout iteration was developed that incorporates the optional functional blocks, including the LED follower stage and the additional amplification circuitry. This expanded version enhances the system's capabilities while maintaining the same grounding strategy, routing principles, and mechanical constraints as the base design. The optional blocks are positioned to minimize interference with the core signal path and to clearly distinguish auxiliary features from critical processing stages. This modular approach provides flexibility in implementation, allowing the system to be deployed in its basic configuration or with extended functionality depending on the requirements of the application.

2.1 Full board

This section presents the full board layout, highlighting the overall component placement, routing organization, and board geometry. This view provides a complete overview of the physical realization of the system.

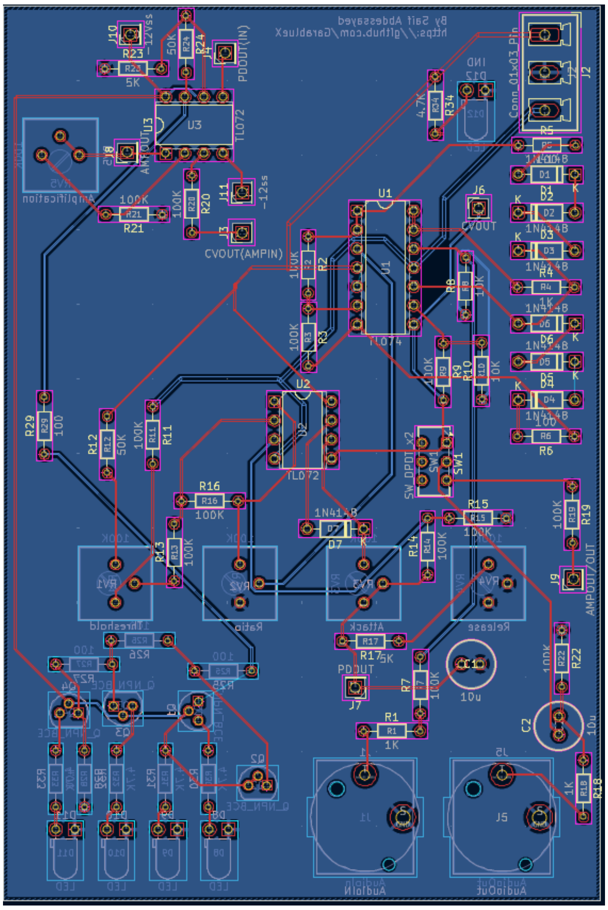

Figure 31: Full Layout

2.2 Top copper

The top copper layer was used primarily for routing signal and power traces. Critical analog paths and feedback loops were kept short and direct, while the ±12 V supply rails were routed with wider traces to ensure reliable power distribution.

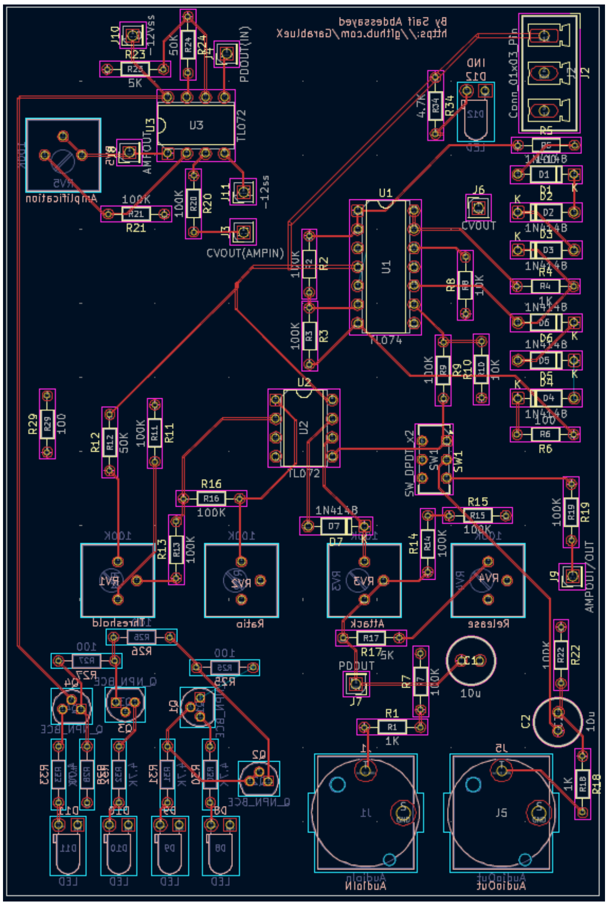

Figure 32: Top Copper Full

2.3 Bottom copper

The bottom copper layer contains the continuous ground plane for the complete PCB. Critical analog paths were kept short and direct on the top layer, and the ±12 V power rails were routed with increased trace widths to ensure reliable operation.

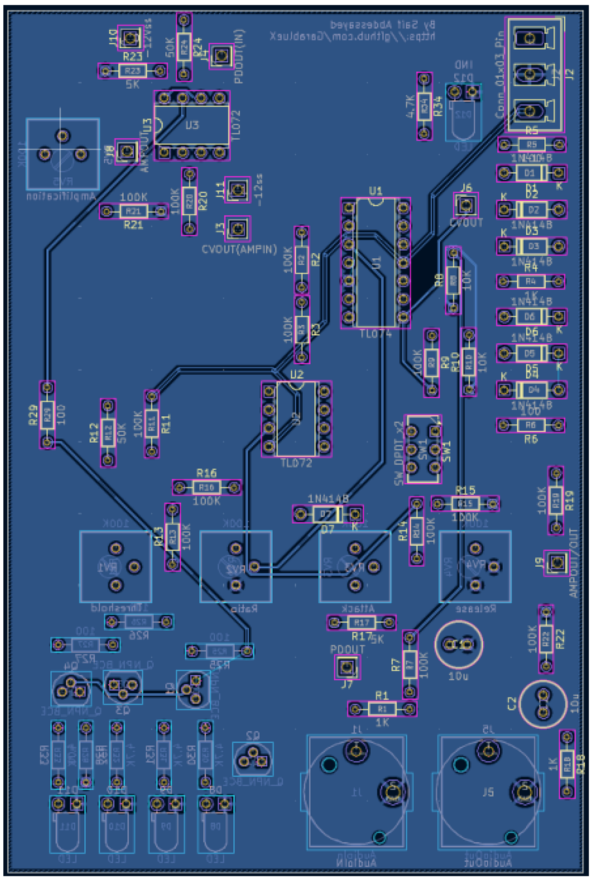

Figure 33: Bottom Copper Full

2.4 3D View

A three dimensional (3D) view of the PCB was generated to verify the mechanical integrity of the design and to validate component placement prior to fabrication.

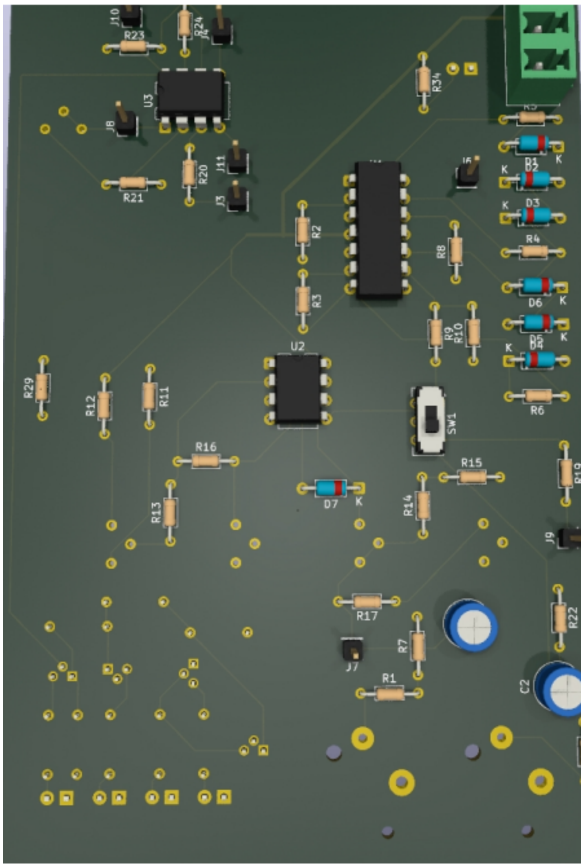

Figure 34: Front Face Full

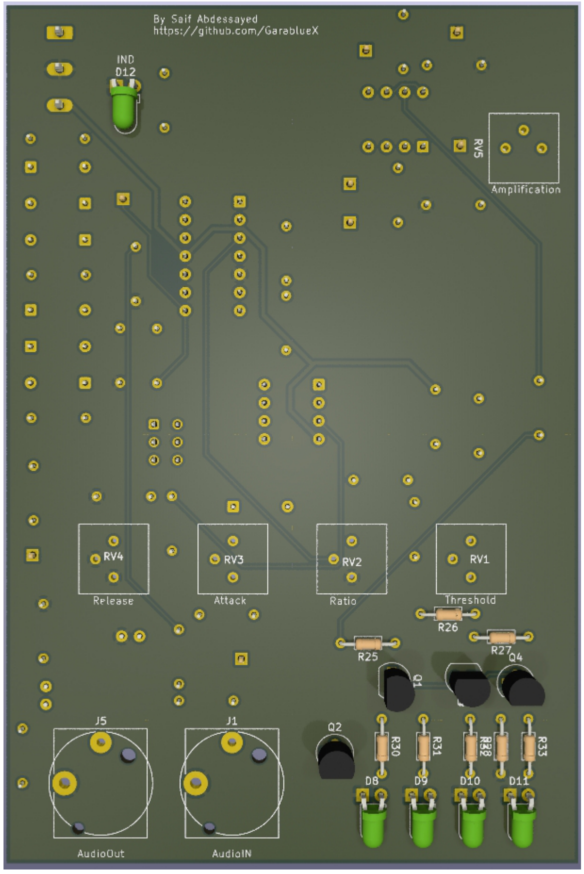

Figure 35: Back Face Full

VII. Conclusion

This project successfully accomplished the design and implementation of an analog signal conditioning system, progressing from schematic conception through to complete PCB layout. Circuit schematic validation, component selection, cost estimation, and power consumption analysis were all carried out systematically. A structured PCB layout was subsequently developed using KiCad, with careful attention to component arrangement, routing strategy, and grounding practices to ensure reliable performance of the analog circuitry.

A modular design approach was adopted by separating the core system from the optional functional blocks, thereby reducing unnecessary complexity for users who require only the fundamental functionality while allowing for implementation flexibility. The final PCB design incorporates a robust ground plane, appropriate trace widths for power and signal routing, and dedicated test points to facilitate debugging and validation.

Overall, the project resulted in a complete, manufacturable PCB design that satisfies the original functional requirements and can be directly fabricated or further expanded upon. Potential future work includes physical prototyping, experimental validation, and enclosure integration.

The following table summarizes the key design parameters that influence the system's dynamic behavior and the associated trade-offs considered during the design process.

Parameter	**Typical Range**	**Design Influence**	**Trade-off Considerations**
Threshold Voltage	0.1 V – 5 V	Compression onset point	Lower vs. higher threshold affects dynamics preservation
Compression Ratio	1.5:1 – 10:1	Dynamic range reduction amount	Higher ratios provide more compression but increase artifacts
Attack Time	10 μs – 100 ms	Transient response speed	Faster attack prevents clipping but may distort transients
Release Time	50 ms – 2 s	Gain recovery duration	Slower release prevents pumping but delays full recovery
Circuit Noise	$<$ −90 dBV	Signal integrity floor	Affects low-level signal preservation

Table 4: Key Design Parameters and Trade off Considerations

Appendix: Development Timeline

The following table presents the development timeline as recorded by the project's version control history, illustrating the progression from initial circuit design through to final deliverables.

Date	Phase	Activity
Dec 15–20	**Research**	Literature review, system architecture definition
Dec 21–25	**Research**	Component selection and topology study
Dec 26–28	**Circuit Design**	Voltage control block design (diode ladder)
Dec 29	**Circuit Design**	Peak detection block design
Dec 30	**Circuit Design**	VC block modifications and simulations
Dec 30	**Circuit Design**	Peak detector fixes and simulation validation
Dec 31	**Circuit Design**	Comparator block and system assembly
Jan 5	**Features**	Attack, Release, and Ratio controls
Jan 6	**Features**	Gain control and amplification stage
Jan 7	**Features**	LED follower block
Jan 9	**Features**	Input voltage clamping
Jan 10	**CAD Migration**	KiCad schematics migration
Jan 11	**CAD Migration**	Footprints, 3D models, schematic refinement
Jan 11	**Analysis**	Power consumption analysis
Jan 12	**PCB Layout**	Initial PCB layout
Jan 14	**PCB Layout**	Layout and routing process
Jan 15	**PCB Layout**	First layouts and 3D views
Jan 16–18	**PCB Layout**	Final layout revisions and assembly files
Jan 19–31	**Finalization**	Repository cleanup, README, final revisions
Feb 1–4	**Finalization**	Final file updates and review
Feb 5–10	**Documentation**	Report writing, formatting, and compilation

Table 5: Development Timeline Extracted from Version Control History

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